



Ankit Kumar Singh
Roll No.:220107012
B.Tech - Chemical Engineering
Indian Institute Of Technology, Guwahati

+91-7800718993
ankitk.snh@iitg.ac.in
ankitektasingh@gmail.com
[Github](#) | [LinkedIn](#)

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.65*	2022-Present
Senior Secondary	CBSE Board	95.6%	2021
Secondary	CBSE Board	94.8%	2019

EXPERIENCE

- **Reliance Industries Limited**

May 2025 - Jul 2025

GET Summer Intern, DTA Hydrotreater Plant
Jamnagar, Gujarat

– Developed and implemented a simulation model using **HTRI** to analyze **fouling trends** in the heat exchanger system

– Derived data-driven insights from simulation output, supporting the **predictive maintenance** and system optimization

– Analyzed fouling trends over **3.5 years**, identifying significant **64% decline** in heat duty before the scheduled cleaning

– Optimized performance by analyzing outputs, collaborating with engineers, and recommending **mitigation strategies**

PROJECTS

- **Credit Risk Financial Analysis**

Mar 2025 - Apr 2025

Self Project
[bit.ly/Credit_Risk](#)

– Performed **EDA** on **32,000+** applicants using Univariate and Bivariate analysis to detect outliers and visualize key features

– Preprocessed data using **KNN Imputer** and **SMOTE** technique, improving minority class recall by **26%** across models

– Developed and compared ML models, with **Random Forest** achieving **93%** accuracy and **0.93 F1** for credit risk prediction
- **Urban Retail Co. Inventory Management**

May 2025 - Jun 2025

Consulting and Analytics Club, IIT Guwahati
[bit.ly/Urban_Retail](#)

– Analyzed **100K+** inventory records using SQL queries to calculate stock levels, estimate reorders, and evaluate turnover

– Developed an **ERD** for retail inventory data, enabling scalable analysis across **5** regional warehouses and **5000+ SKUs**

– Created interactive **Tableau** dashboards and detailed **KPI** reports, improving data visualization and operational efficiency
- **Bubble Size Prediction using ML**

Mar 2025 - May 2025

Course Project under Prof. Senthilmurugan Subbiah
[bit.ly/Bubble_size](#)

– Cleaned and analyzed **1,500+** data points with key preprocessing steps like **normalization**, **outlier removal**, and **PCA**

– Tested five machine learning models, achieving **R² 0.981** and **RMSE 0.0049** with **CatBoost** using categorical features

– Deployed the best-performing model via an interactive **Streamlit** web app for seamless real-time bubble size prediction

TECHNICAL SKILLS

- **Programming:** C/C++, Python

• **Data Analysis:** MySQL, MS Excel

• **Data Visualization:** Tableau, Matplotlib, Seaborn

• **Python Libraries:** Numpy, Pandas, Scikit-learn

KEY COURSES TAKEN

- **Data Science:** SQL for Data Science by UC Davis, Google Data Analytics
- **Machine Learning:** Machine Learning by DeepLearning.AI, Application of AI and ML in Chemical Engineering
- **Mathematics:** Linear Algebra, Basic Calculus, Real & Complex Analysis, Partial Differential Equations
- **Computer Science:** Data Structures and Algorithms, Python for Everybody Specialization, Introduction to Computing

ACHIEVEMENTS

- **ISM School of Business, Lithuania | Creative Shock:** Social Business Case, **International rank 44** among **1500+** teams
- **StartUproar:** Distinguished as a **finalist** among **200+** teams from top-tier institutions in the Bizthon Ideathon
- **HMEL's QUEST 2025:** Recognized as a **finalist** in a national-level energy and sustainability innovation challenge
- **Winter Consulting 2023:** Ranked in the top **10** percentile among all registered participants.
- **JEE Advanced 2022:** Secured an All India Rank of **5821** among **1,60,000+** students
- **JEE Mains 2022:** Achieved a percentile of **99.24** among **905,590** candidates

POSITIONS OF RESPONSIBILITY

- **Senior Executive**, Entrepreneurship Cell, IIT Guwahati

Apr 2023 - Mar 2024

– Led a team of **15** junior executives in creating social media campaigns, driving **3x YoY** growth and **1.5x** participation

– Organized annual Scavenger Hunt for **900+** freshers, managing **30+** volunteers, event flow, marketing and content
- **National Entrepreneurship Drive (NED) Manager**, UDGM 2024, IIT Guwahati

Oct 2023 - Jan 2024

– Founded National Entrepreneurship Drive, established **20+** E-Cells | Organized Startup Expo for **40+** Northeast startups

– Organized a **4-week** long cohort from scratch in colleges all over India, to build their E-Cells